Requirements

Functional Requirements

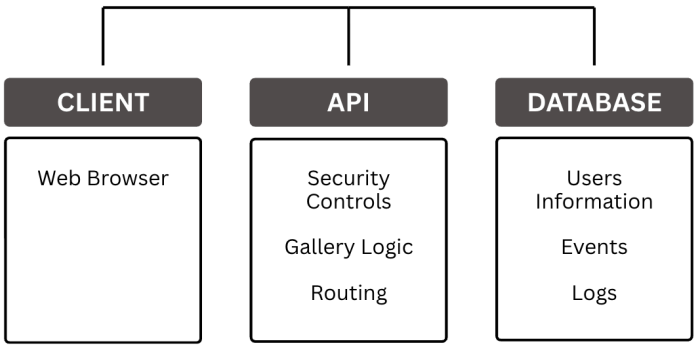
- All users can enter/leave galleries and rooms, given they're not full
- All users can see room occupancy
- Users with higher access can view and edit more information

Security Requirements

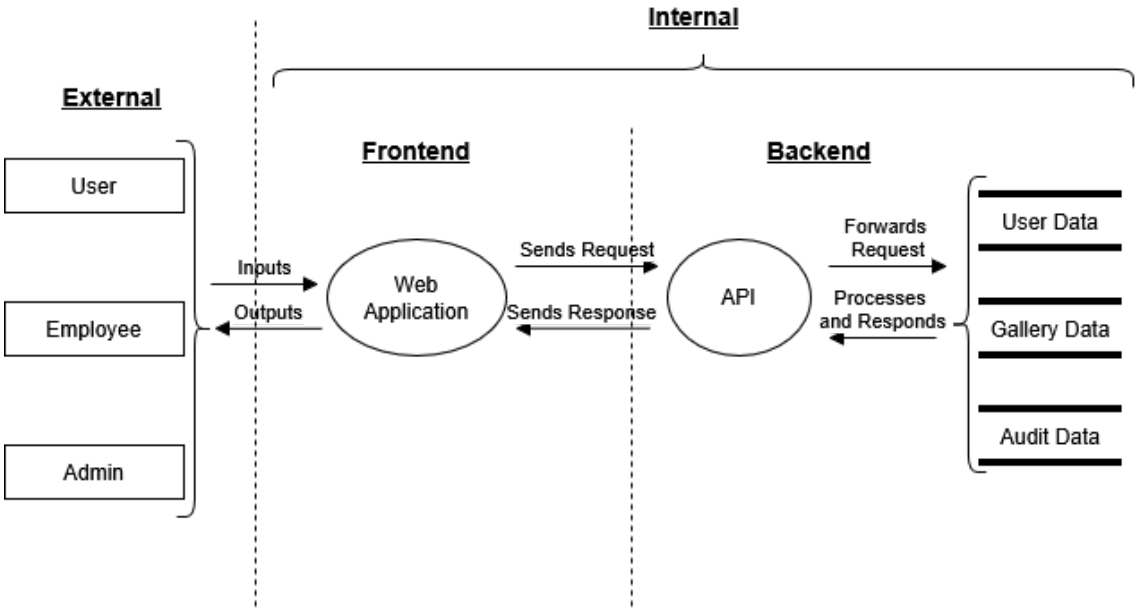
- Passwords are stored securely
- All information is validated before it's stored/entered
- Vulnerabilities are mitigated
- Error handling practices are implemented (ex., leaving a room you aren't in)

Diagrams

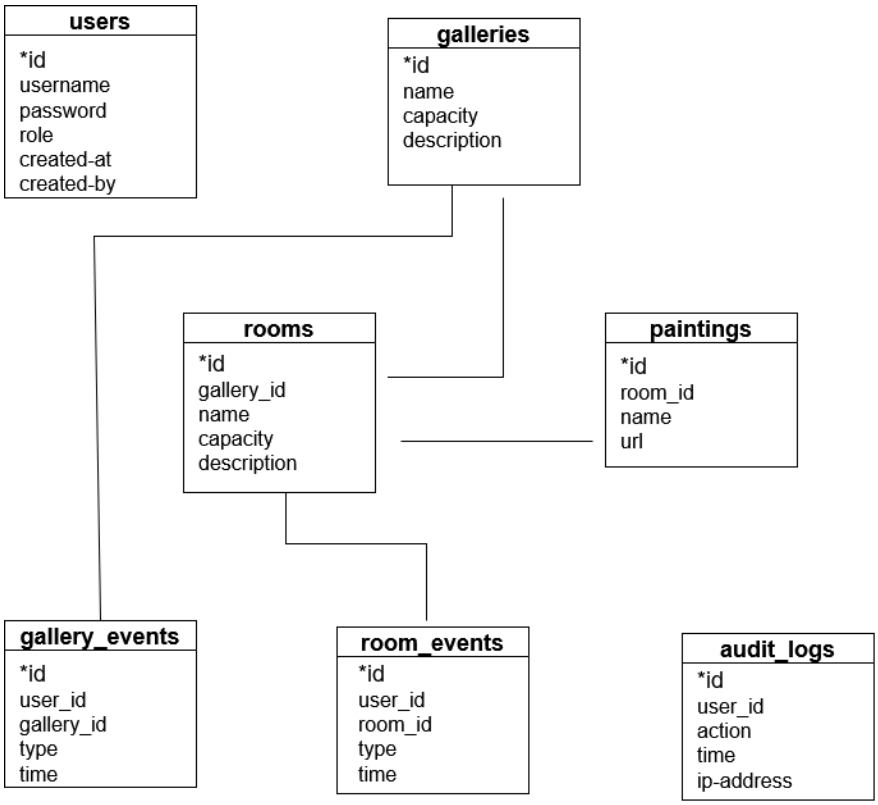
Architecture Diagram



Data Flow Diagram



ER Diagram



STRIDE Threat Model

Threat	Component	Attack Scenario	Impact	Mitigation
Spoofing	Login	Attacker obtains another user's credentials and logs in as them	Exposure of protected data	Password hashing
Tampering	Database	Attackers use SQL injections to impact data in the database	Data modification	Input validation, server-side authentication
Repudiation	Gallery/Room Events	A user claims they never entered a room/gallery	Validity of user actions is questioned	Record information about gallery/room events (time it happened, who the user was, etc)

Information Disclosure	API	A user uses a known path to access information they normally don't have access to view	Exposure of protected data	Authorization and authentication
Denial of Service	Login	Attacker floods the server with login requests	Service becomes slow and/or inaccessible	Rate limiting
Elevation of Privilege	Database	An attacker modifies their account role to gain higher access	Unauthorized database access and data modification	Implement access controls